



Incorporating Big Data in Statewide and MPO Travel Demand Models in Tennessee

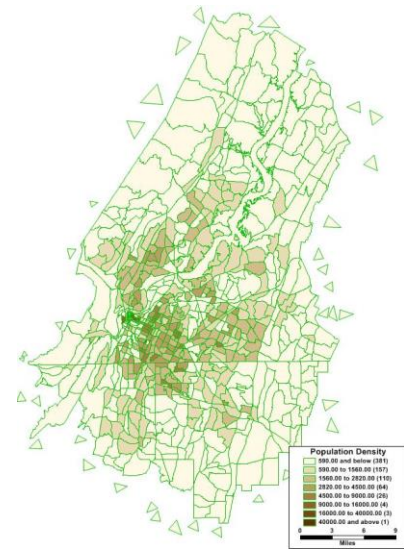
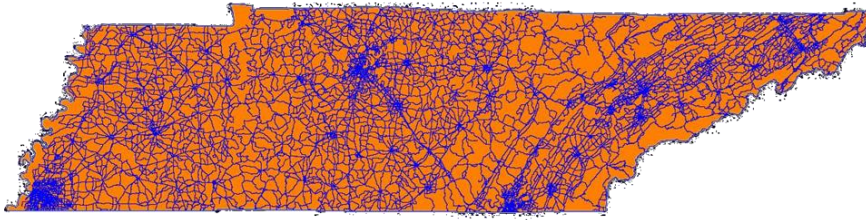
Vince Bernardin, PhD, RSG

December 8, 2016

What does Big Data mean for modeling?



Recent Experiences

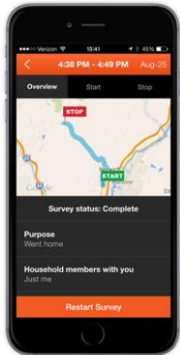


- Tennessee Statewide Model
 - First statewide model calibrated with AirSage
- Chattanooga MPO Model
 - First activity-based model incorporating AirSage

rMOVE
by RSG

- Other experiences

- Ohio (US 30 study with ATRI data)
- California, New York, Indiana, Michigan, Virginia, Maryland, North Carolina, Florida, Iowa, Utah, Idaho, Alaska...
- rMove, Bluetooth





The Power of Big Data

The Power of Big Data

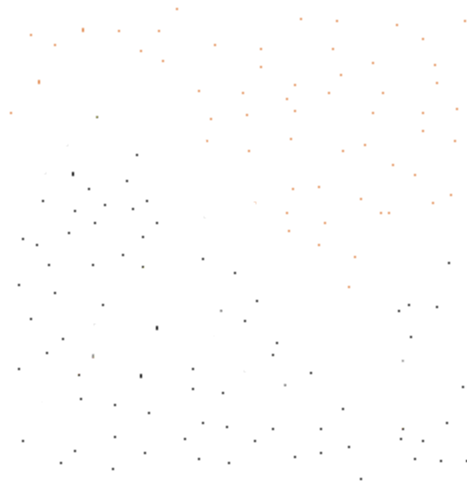
TN STATEWIDE DATA

- Combined household survey
 - NHTS + 4 MPOs
 - 10,344 households
- AirSage and ATRI datasets
- Trip Table (OD pairs)
 - Total: 12,744,900
 - Survey: 39,782 0.3%
 - AirSage: 3,355,539 26.3%

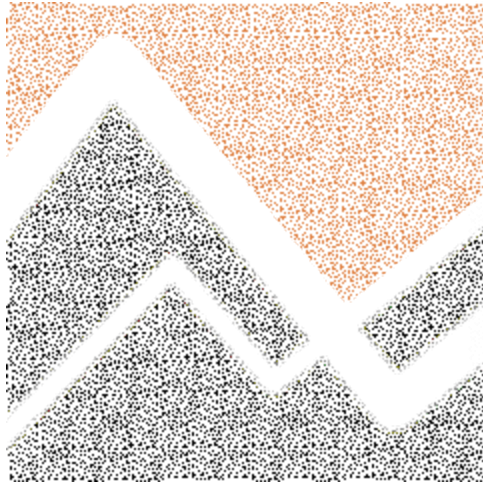
CHATTANOOGA DATA

- 2010 household survey
 - 1,502 households
- AirSage and ATRI datasets
- Trip Table (OD pairs)
 - Total: 529,984
 - Survey: 8,350 2.0%
 - AirSage: 182,742 34.5%

Can you recognize the pattern based on $\leq 2\%$?



How about based on >25%?



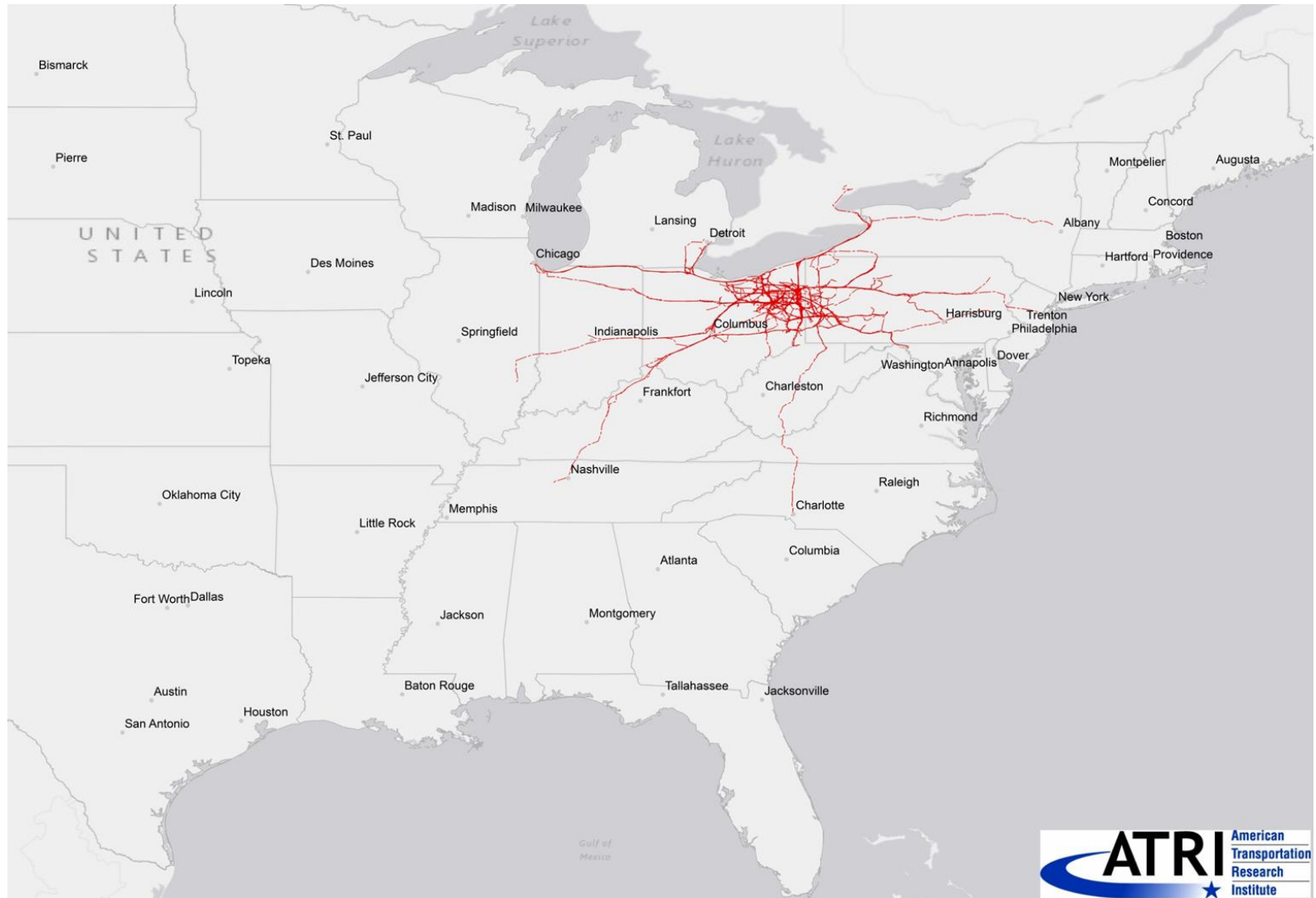
Big Data allows us to see the Big Picture



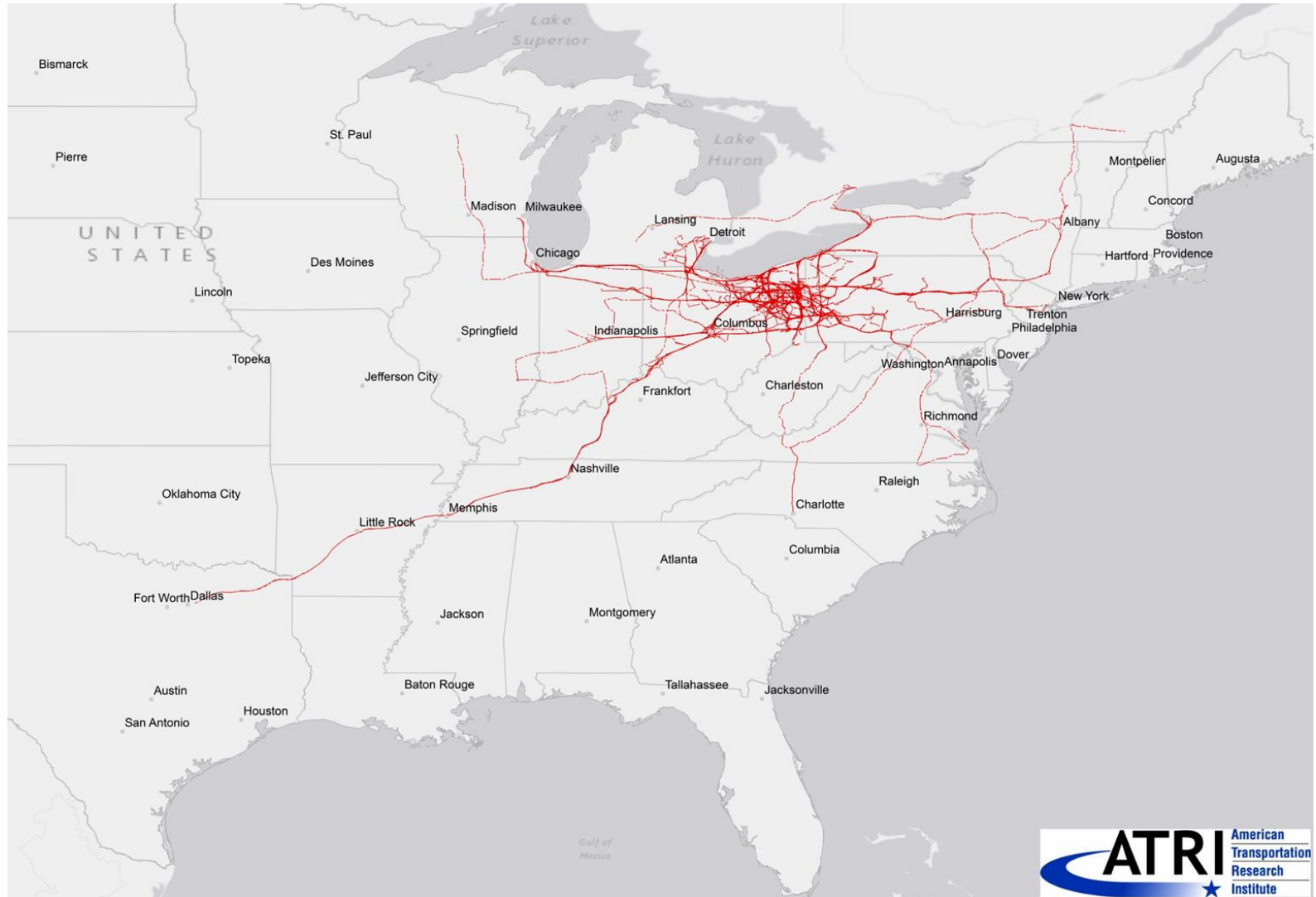
US 30 Study Area



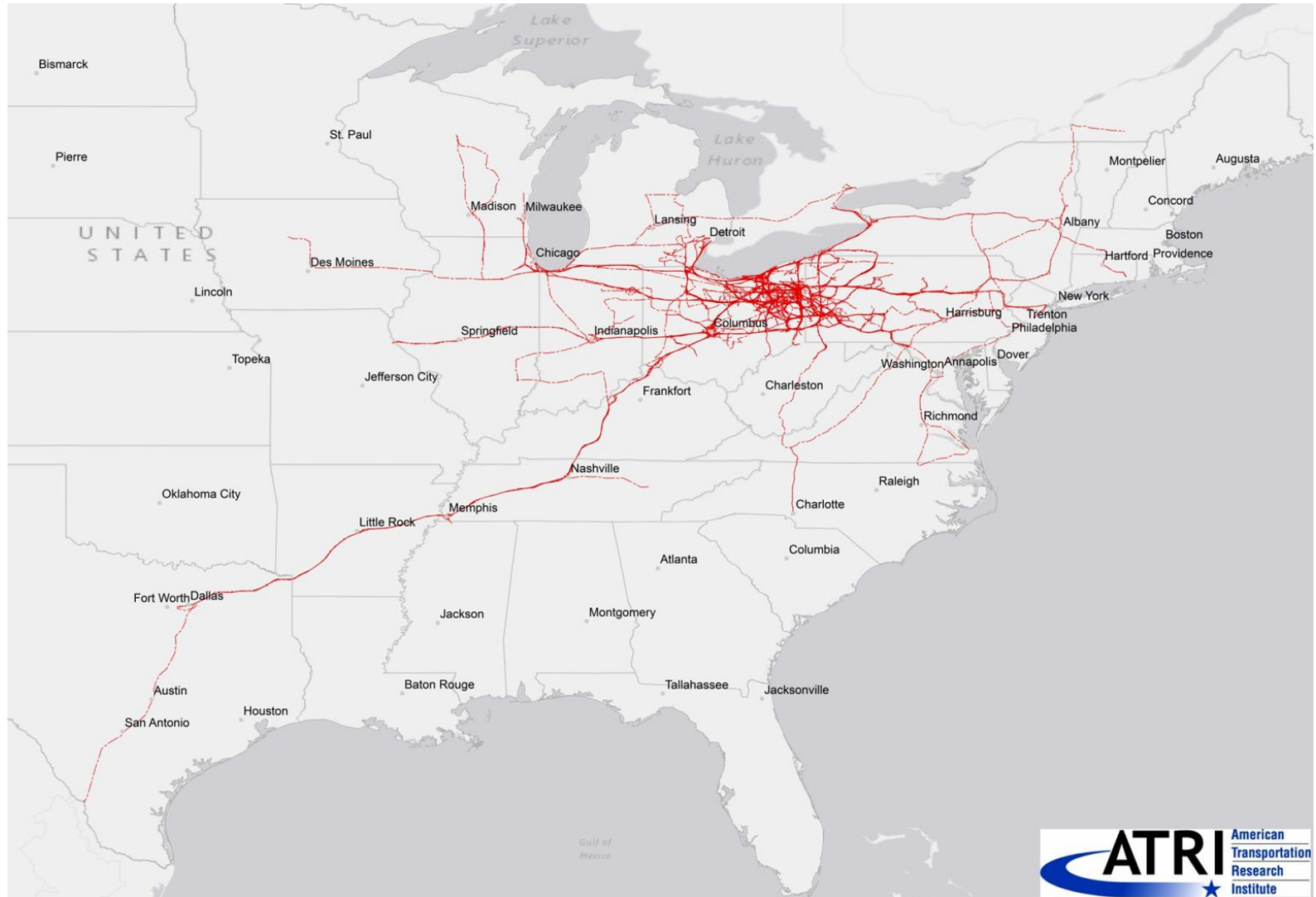
Trucks using the US 30 corridor – after 1 Day



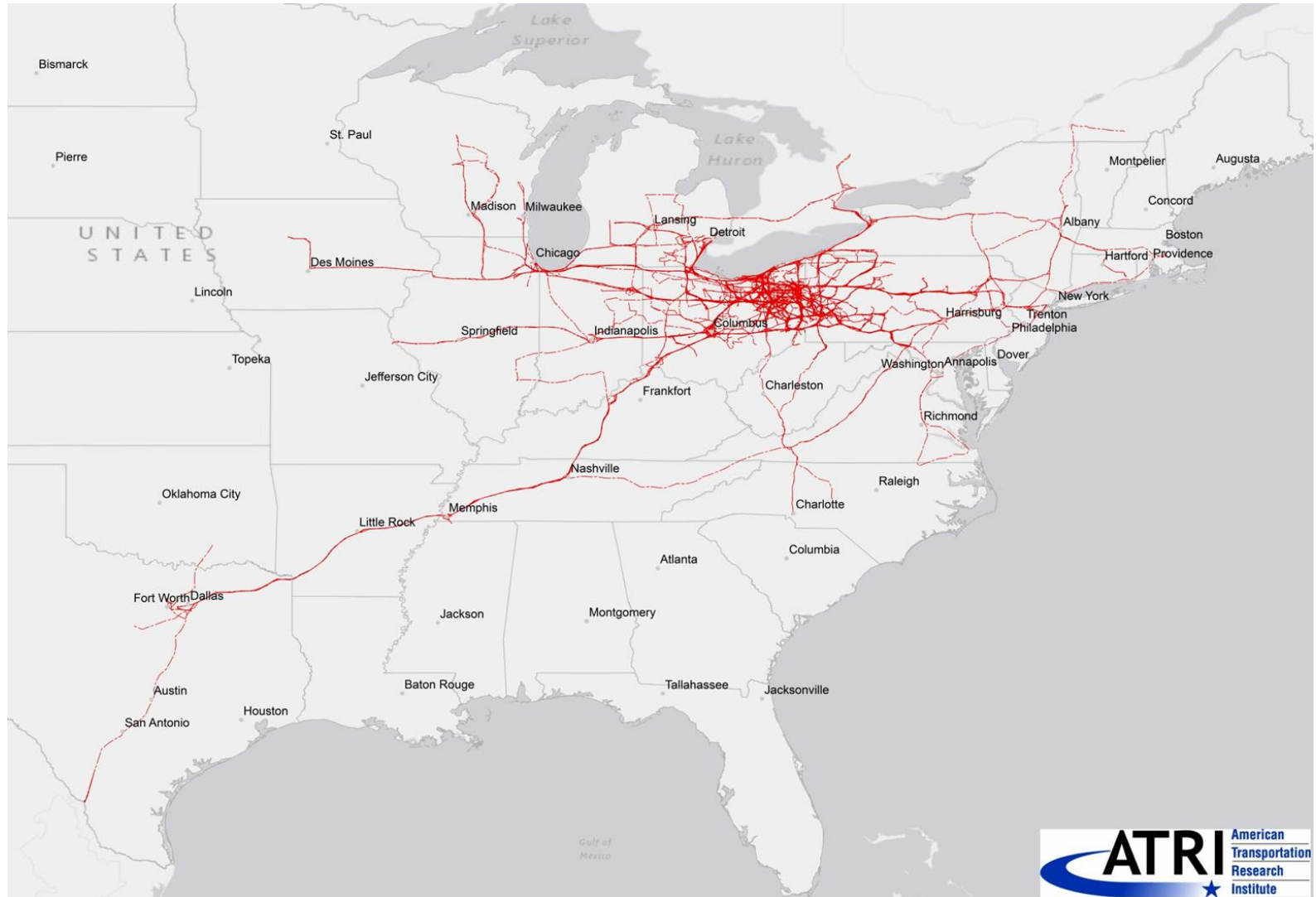
Trucks using the US 30 corridor – after 2 Days



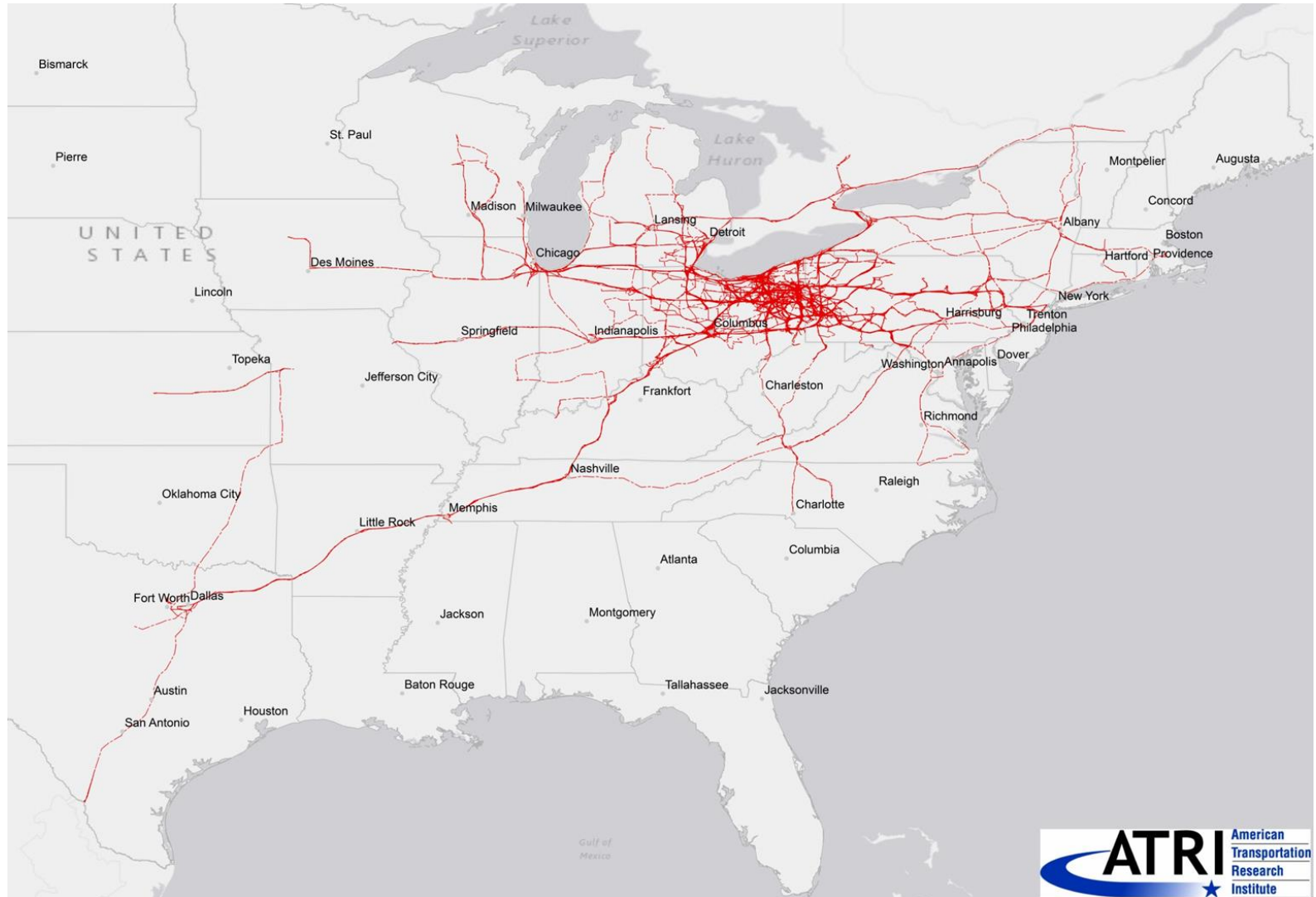
Trucks using the US 30 corridor – after 3 Days



Trucks using the US 30 corridor – after 5 Days



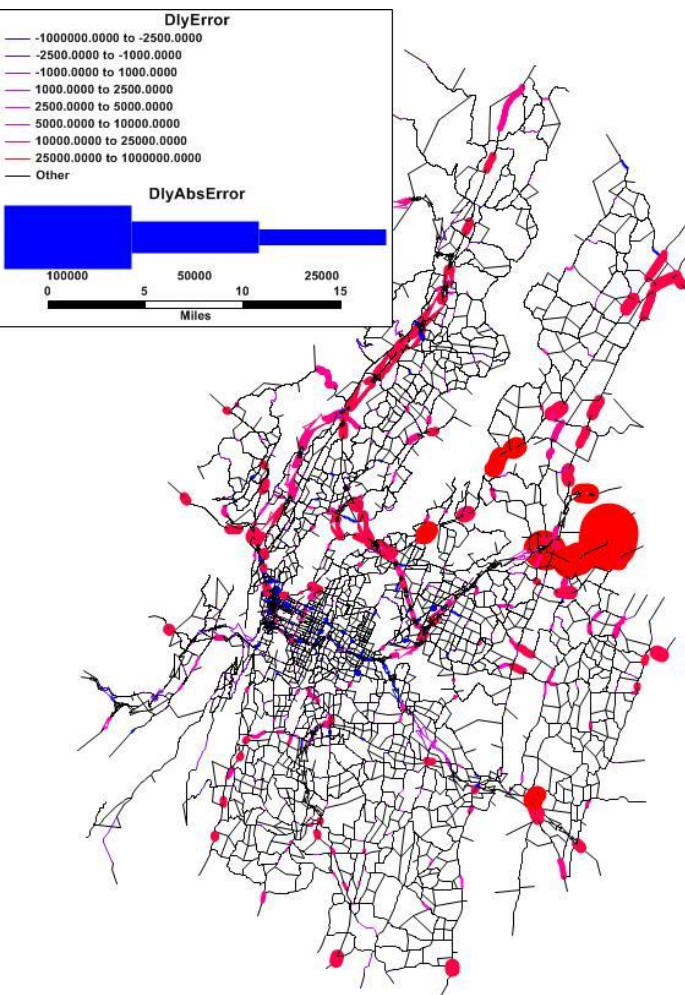
Trucks using the US 30 corridor – after 7 Days





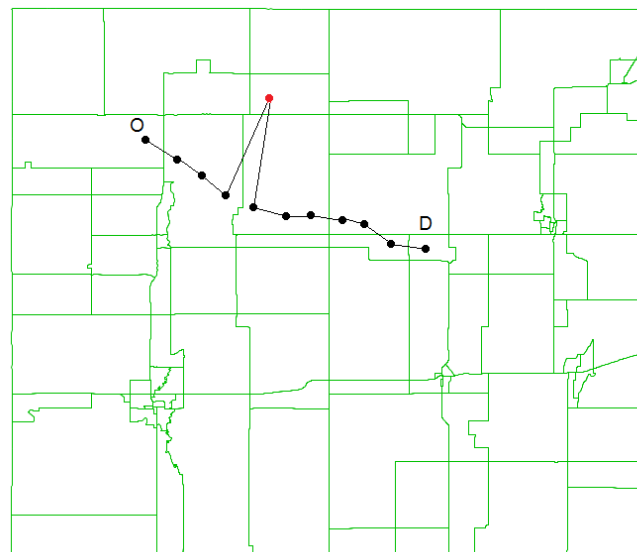
The Limitations of Big Data

Cleaning Required

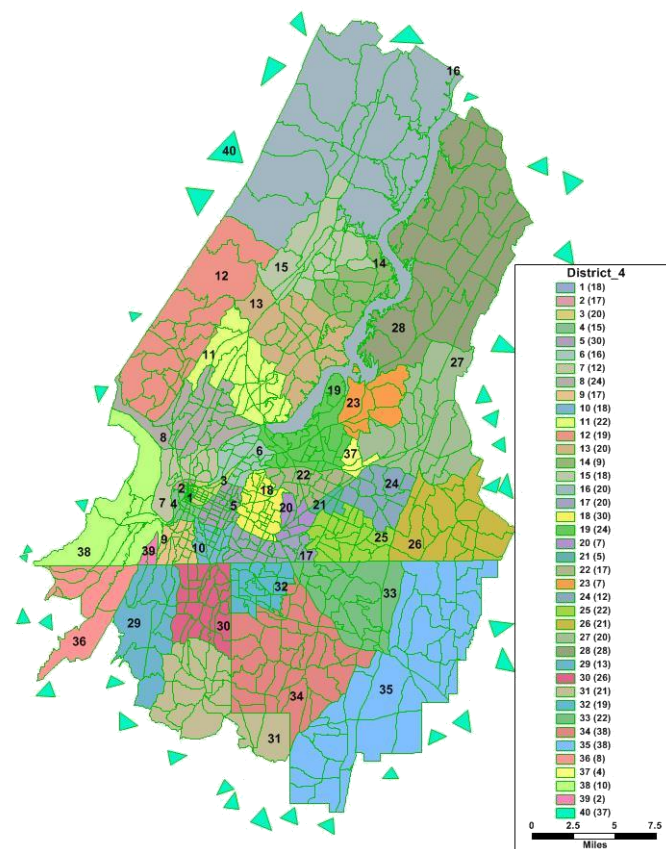
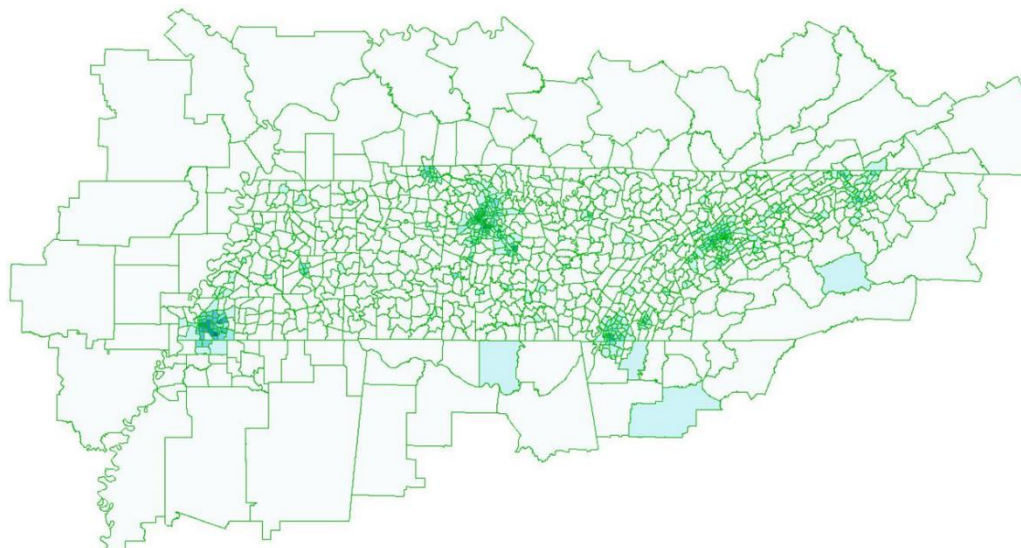


- Filtering / cleaning

- Needs vary by data source – but all need it
- GPS jumps/blips and equivalent
- Missing data



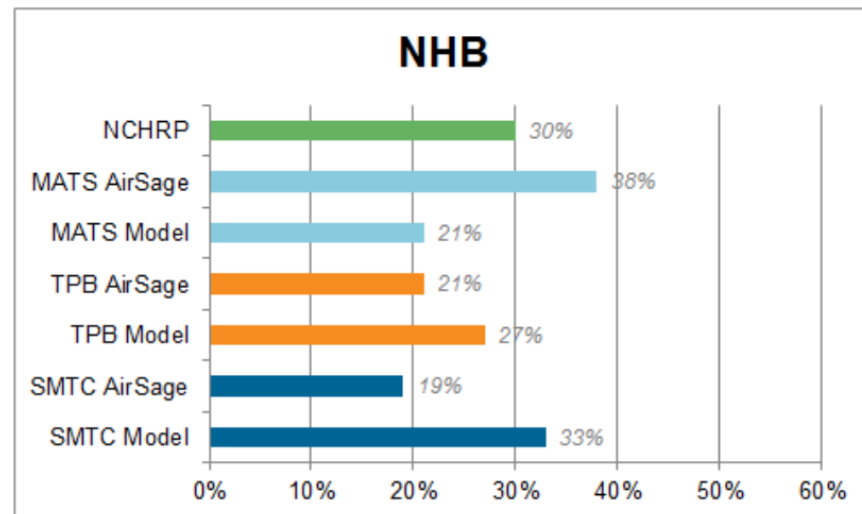
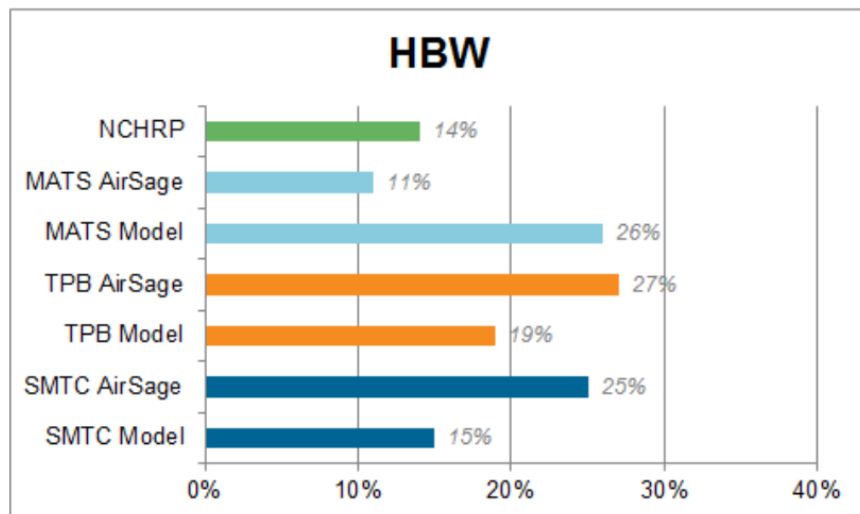
Limited Spatial Resolution



- Accuracy and Cost
 - Cell-based has limited precision
 - Pricing based on number of zones / districts

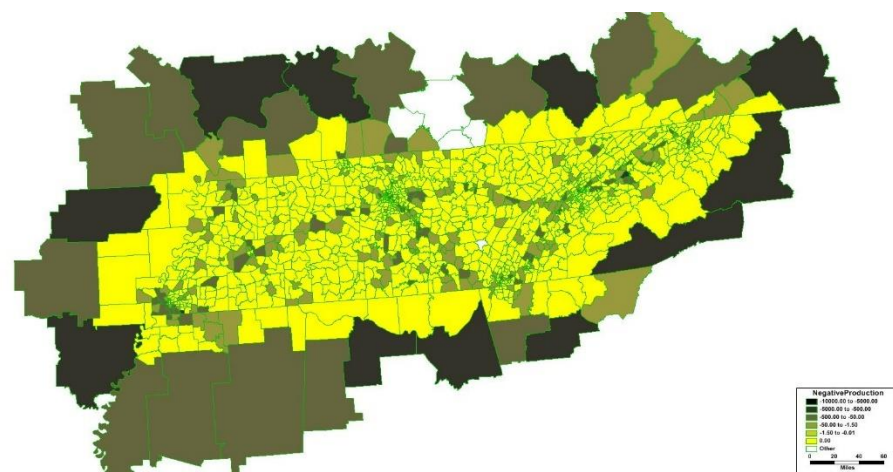
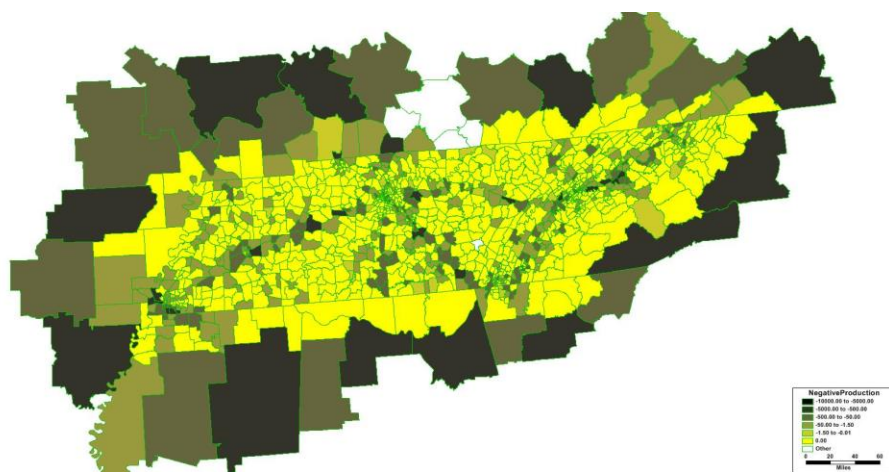
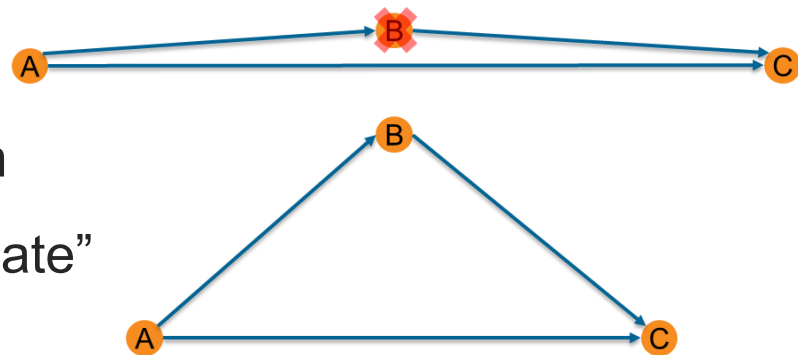
No Purpose or Mode

- Just ODs – not a survey substitute
 - Imputation can't reproduce surveys (so far)
 - GPS ID persistence prohibitive
 - Better to supplement with CTPP / LEHD



Trip Definitions

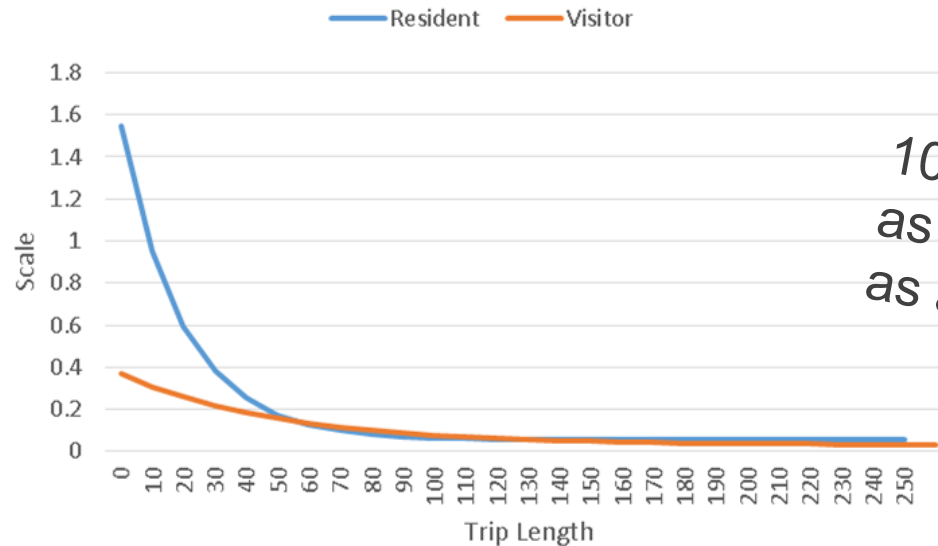
- Combining AirSage and ATRI
 - Need consistent trip/stop definition
 - Whether or not to count “intermediate” stops and break up long trips



- Filtering short stops eliminated 87% of inconsistencies
 - From 11% of cells & 0.20% of trips to 1% of cells and 0.09% trips

Not Representative

- Big Sample NOT Random Sample
 - Locational biases, holes
 - Trip length / duration biases
 - Not corrected by penetration-based expansion



*100 mi trip is 12 times
as likely to be detected
as a 10 mi trip*

Data Expansion

- Traffic Counts are unbiased – expand to counts

- ODME-based

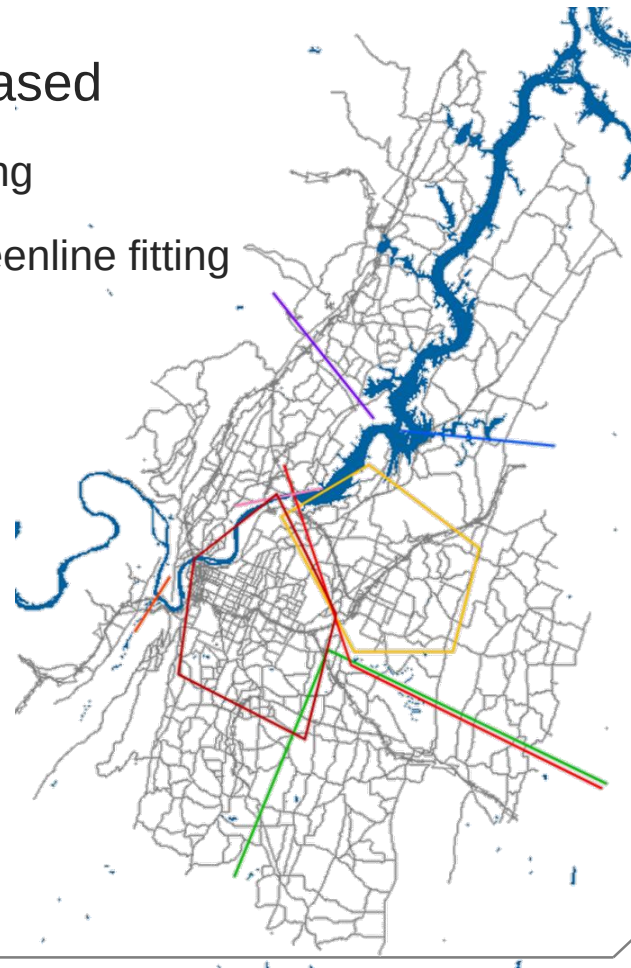
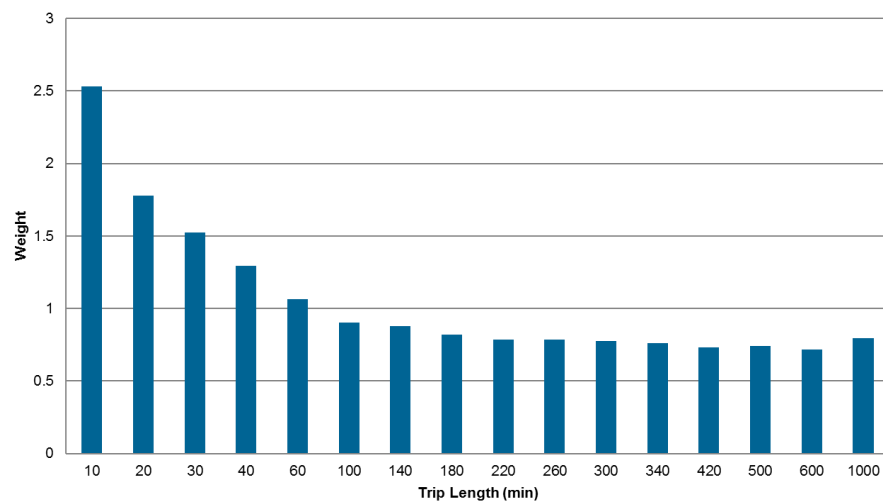
- Direct ODME

- In-direct ODME

- Non-ODME-based

- Simple scaling

- Iterative screenline fitting

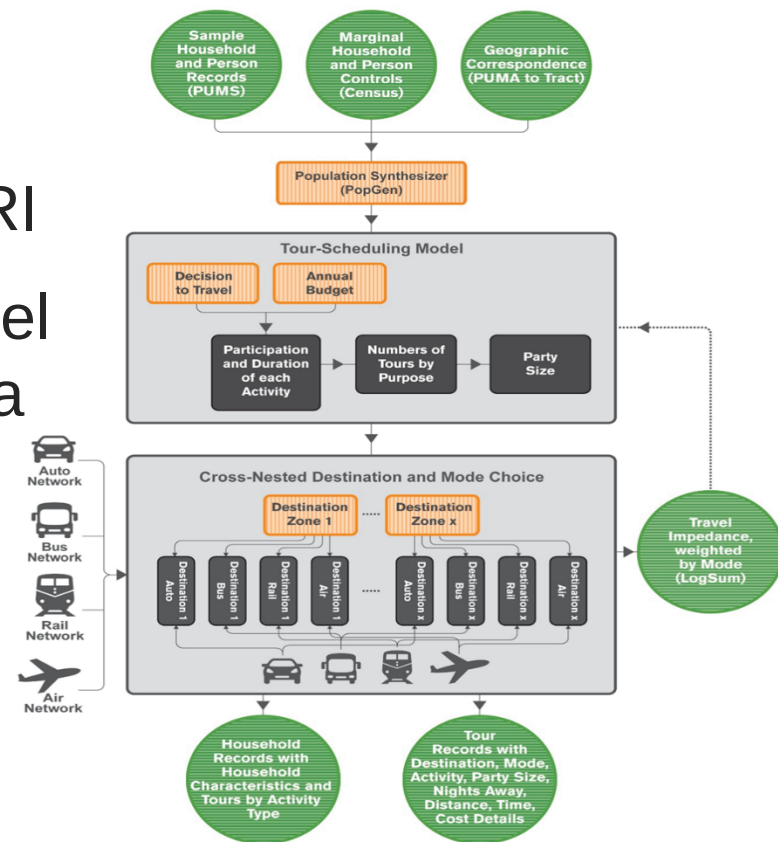
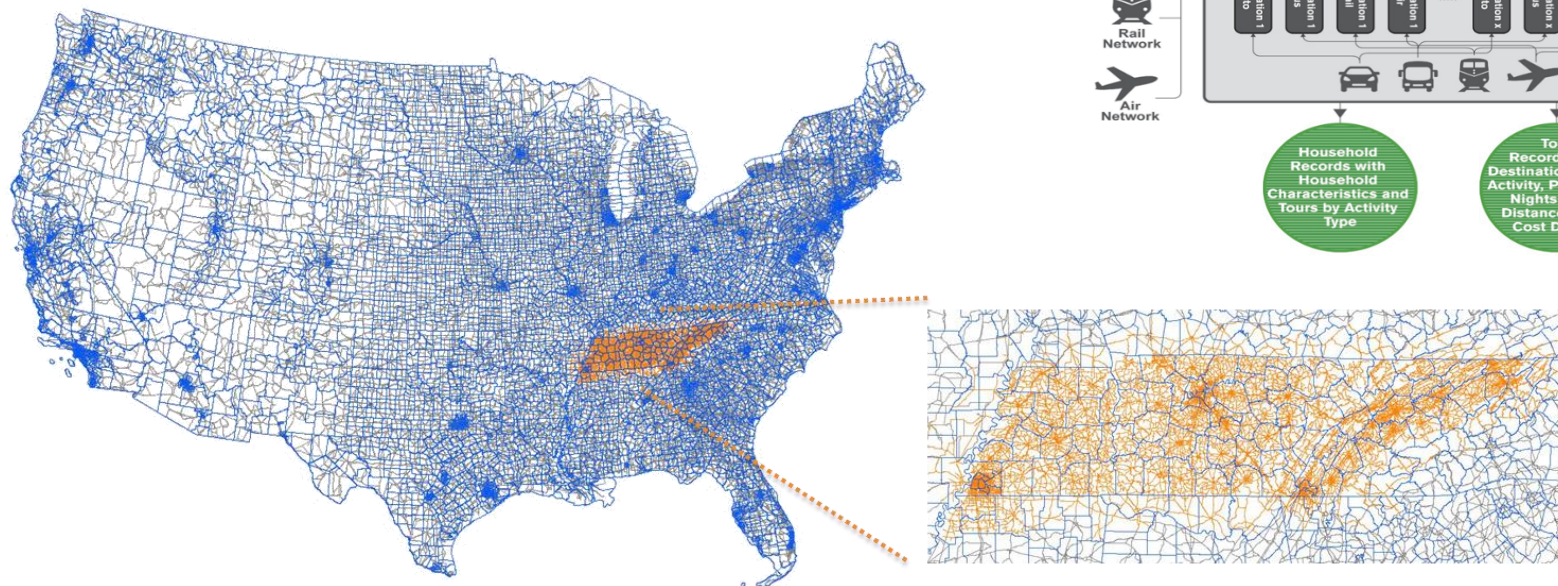




Model Results with Big Data

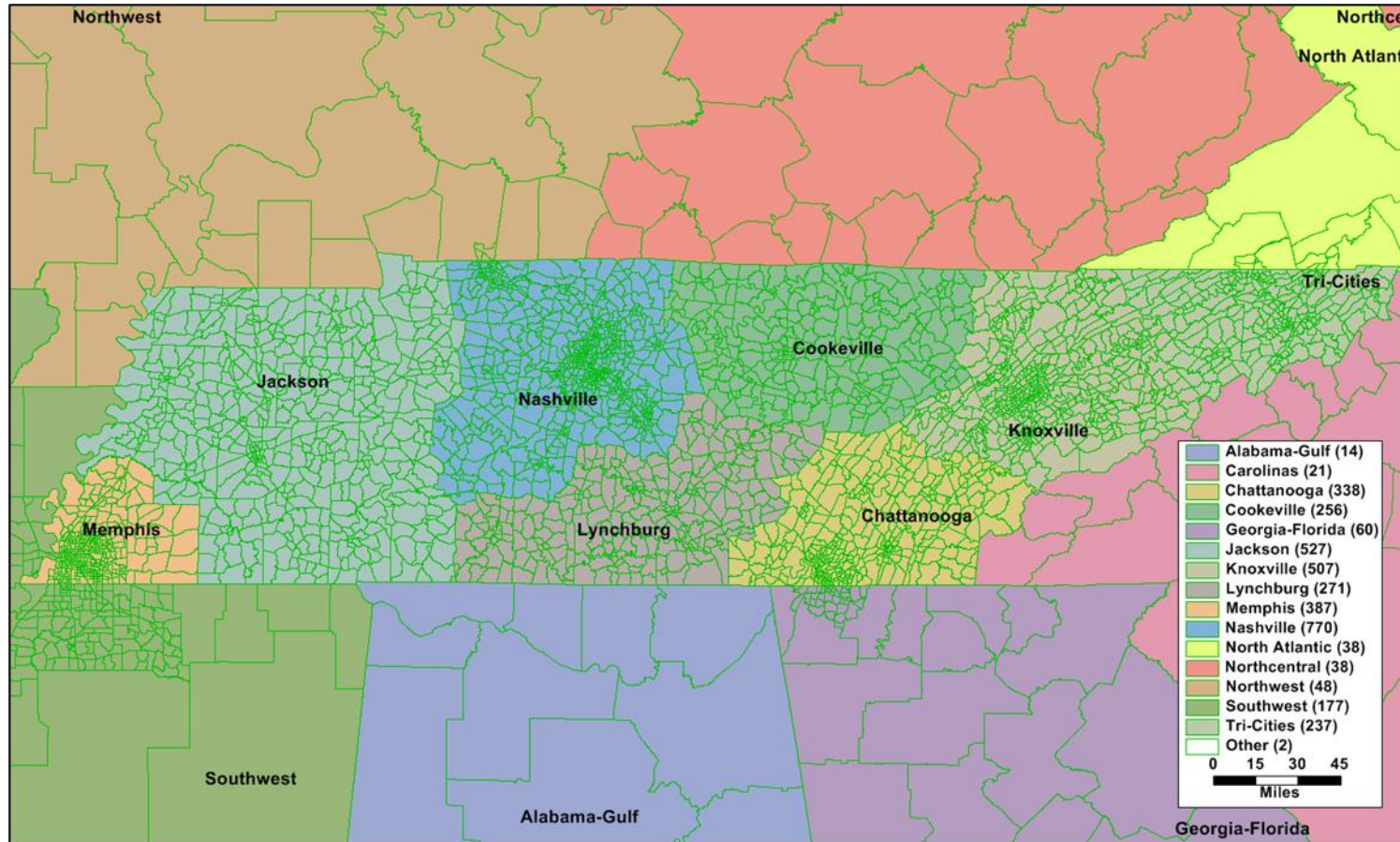
TN Long Distance Trips

- Modeled Trips pivot off AirSage/ATRI
- FHWA National Long Distance Model Calibrated to Regional AirSage Data
 - Psychological bias against state border crossings



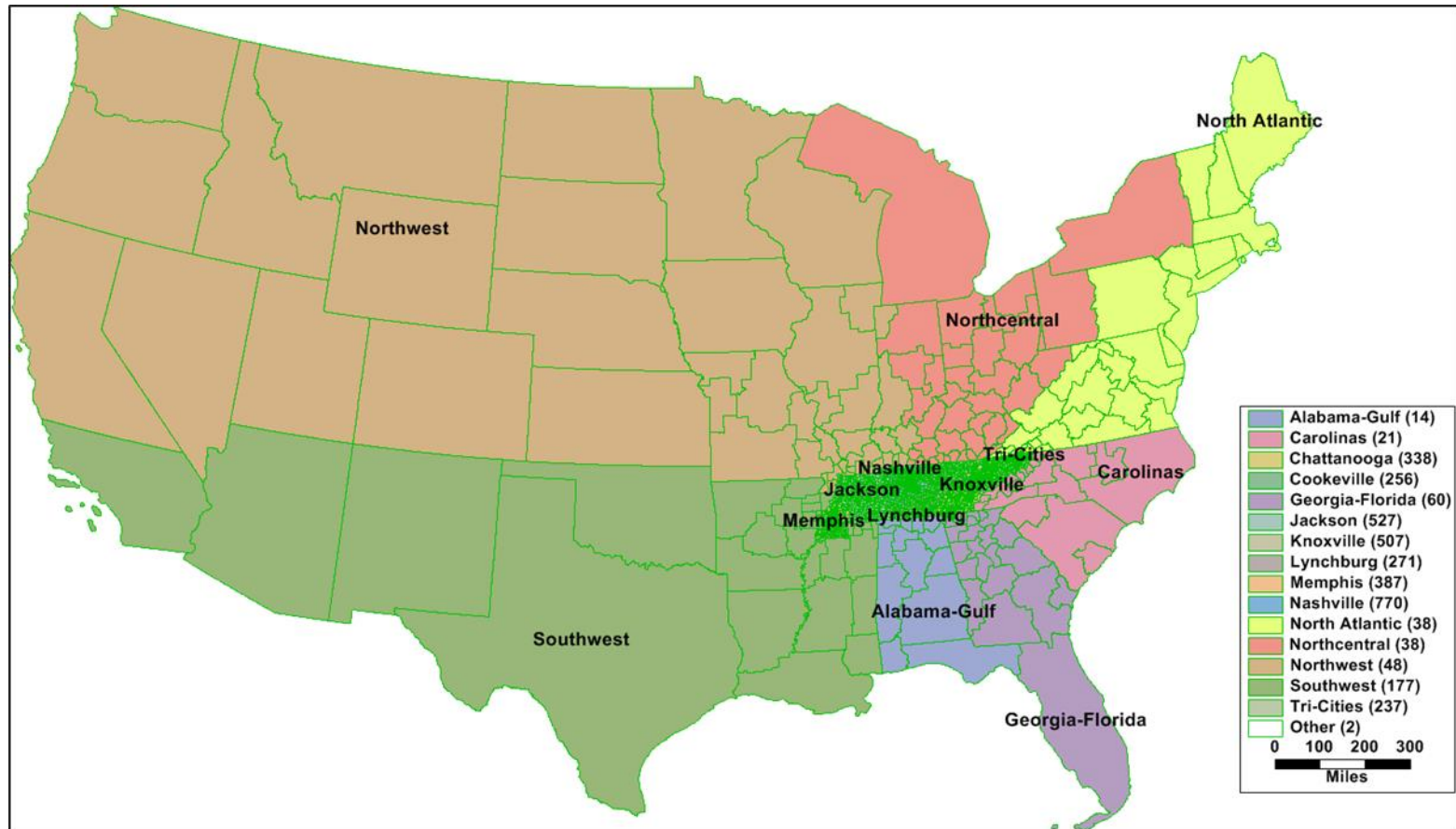
TN Internal Districts

DISTRICT SCHEME (INTERNALS)



TN External Districts

DISTRICT SCHEME (EXTERNALS)



To/From TN Trip Distribution

DISTRICT-TO-DISTRICT COMPARISON

- Generally good agreement
- District level origins & destinations all within 10%, most within 3%
 - Smoky Mtns not attracting enough to/from Knoxville
- District level ODs all within 4% except within Nashville - Northcentral

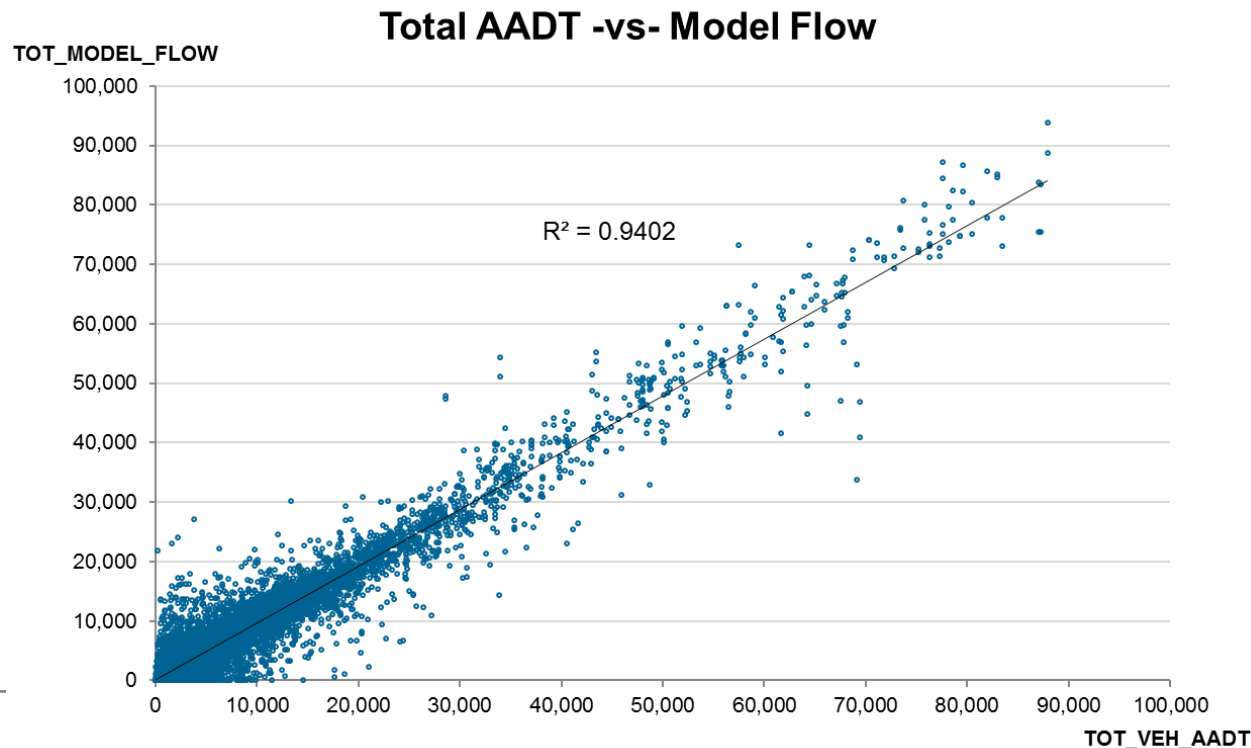
Relative Percentage Difference (Model Version 3 vs AirSage) I-E & E-I Trips

Internal districts	External districts							Total
	Northwest	North Atlantic	Northcentral	Carolinas	Alabama-Gulf	Southwest	Georgia-Florida	
Tri-Cities	0.4%	0.1%	0.8%	3.6%	0.0%	0.2%	0.3%	5.3%
Knoxville	0.5%	-2.6%	-1.2%	-1.7%	-0.7%	0.3%	-2.0%	-7.3%
Chattanooga	0.0%	-0.1%	-0.5%	-0.4%	-1.1%	0.1%	2.7%	0.8%
Cookeville	0.0%	-0.2%	0.9%	-0.3%	-0.1%	-0.1%	-0.2%	0.0%
Lynchburg	-0.4%	0.1%	0.4%	0.0%	0.7%	-0.1%	-0.4%	0.2%
Nashville	-0.7%	-0.3%	6.6%	-0.8%	-3.6%	-2.3%	-2.0%	-3.1%
Jackson	0.0%	0.1%	0.6%	0.0%	0.0%	-1.9%	0.0%	-1.2%
Memphis	0.5%	0.3%	0.8%	0.1%	-0.1%	3.4%	0.3%	5.2%
Total	0.3%	-2.6%	8.4%	0.5%	-4.9%	-0.4%	-1.3%	0.0%

Assignment Validation

- Great fit - One of best statewide models in the country
- Used ODME – with constraints, (some other statewide models do to)

VOLUME RANGE	RMSE	TDOT TARGET
< 5,000	102.1%	101.4%
5,000 to 10,000	35.6%	56.3%
10,000 to 20,000	22.0%	51.4%
20,000 to 30,000	16.4%	35.7%
30,000 to 40,000	14.8%	32.0%
> 40,000	11.1%	12.2%
All	36.6%	60.0%

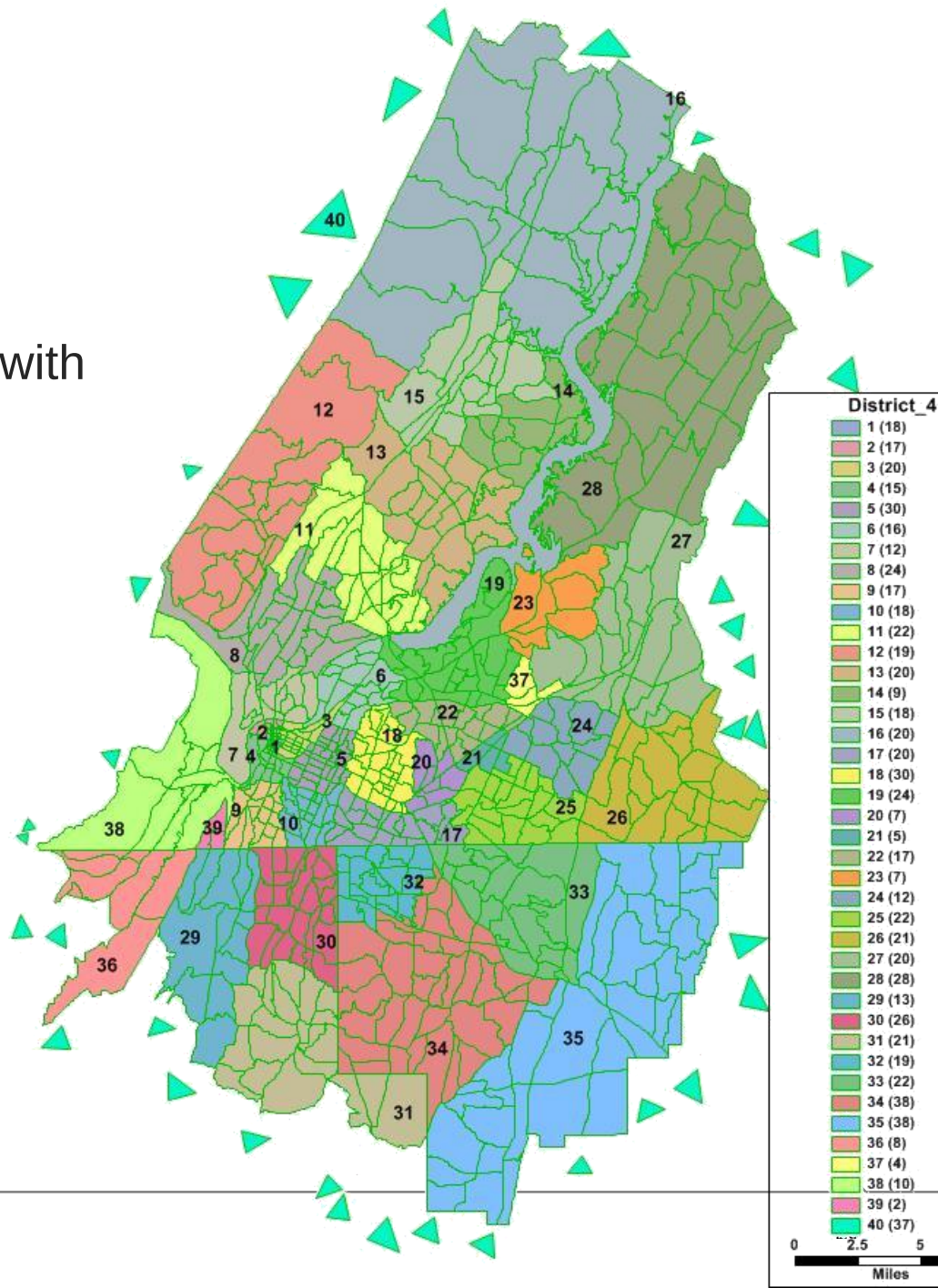


Chattanooga Daysim

- Shadow-Pricing
 - Used 40 district scheme with LEHD and AirSage data

Destination District O-D Shadow Pricing Convergence Summary

Iteration	Absolute Error	Mean absolute % error	Weighted mean absolute % error	RMSE
1	516,595	23.3%	22.2%	37.1%
2	421,404	20.6%	19.1%	30.7%
...	...	...	...	...
24	59,962	11.8%	8.3%	10.5%



Total Daysim Trip Table vs. AirSage

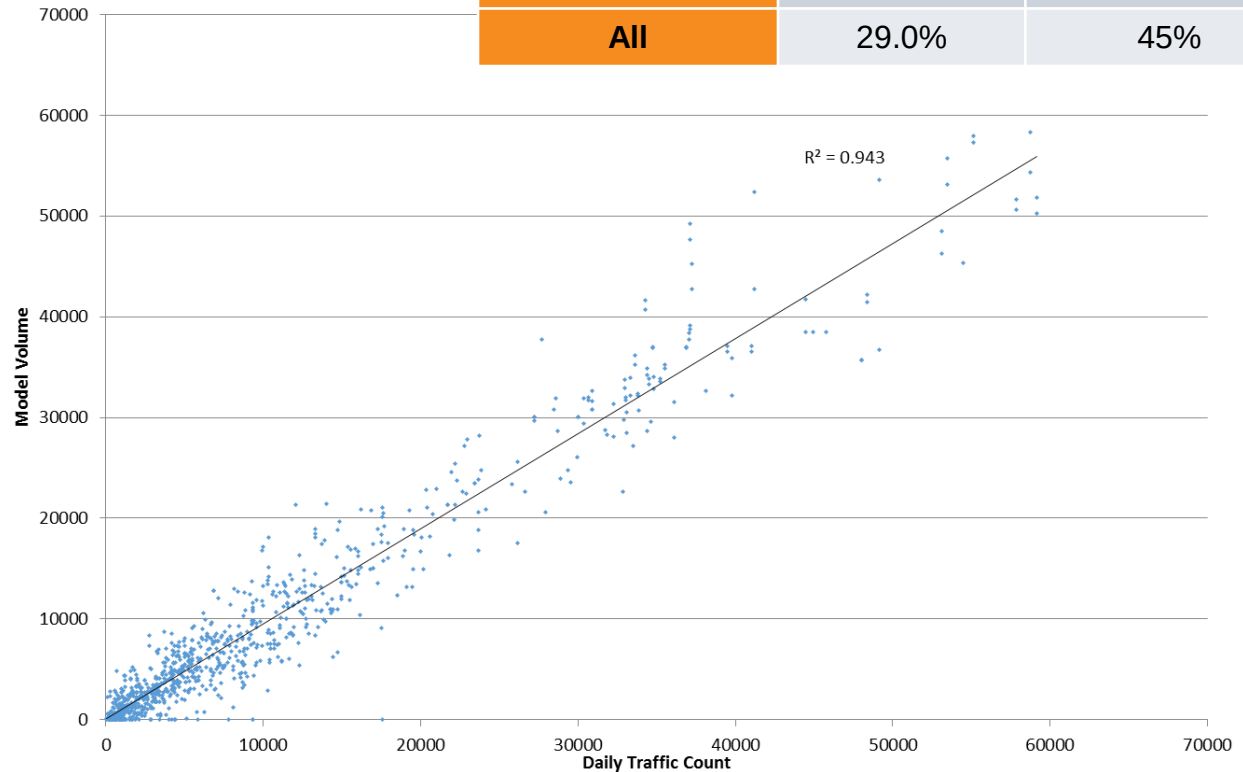
- Daysim vs. AirSage
 - Very good agreement – **10.5% RMSE**
 - All cells within +/- 1%
 - All residence/work Super Districts within +/-2.5%

Origin SuperDistrict	Destination Super District												Grand Total
	1	2	3	4	5	6	7	8	9	10	11	12	
1	0.5%	0.2%	-0.1%	0.0%	0.0%	-0.1%	-0.2%	-0.1%	0.0%	0.0%	-0.1%	-0.2%	0.0%
2	0.3%	0.0%	0.2%	0.0%	0.1%	0.0%	0.0%	0.1%	0.1%	0.0%	0.0%	-0.1%	0.7%
3	-0.1%	0.1%	0.0%	-0.1%	-0.2%	0.0%	0.1%	0.1%	0.0%	0.0%	0.0%	-0.1%	-0.1%
4	0.0%	0.1%	-0.1%	0.0%	0.0%	0.0%	0.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.4%
5	0.1%	0.1%	-0.1%	0.0%	0.2%	0.1%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.5%
6	-0.1%	-0.1%	0.1%	-0.1%	0.1%	0.0%	0.1%	-0.1%	0.1%	0.0%	0.0%	0.0%	0.0%
7	0.0%	0.0%	0.2%	0.1%	0.1%	0.0%	0.2%	0.0%	0.1%	0.0%	0.0%	0.1%	0.7%
8	0.0%	0.1%	0.1%	0.1%	0.0%	-0.1%	0.1%	0.0%	-0.2%	0.0%	0.0%	0.0%	0.2%
9	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.3%	0.0%	0.0%	0.0%	0.2%
10	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.1%	0.0%	0.3%
11	0.0%	0.0%	0.0%	-0.1%	0.0%	0.0%	-0.1%	0.0%	0.0%	0.1%	-0.1%	-0.3%	-0.5%
12	-0.2%	-0.3%	-0.1%	-0.2%	0.0%	-0.1%	-0.2%	-0.1%	-0.1%	0.0%	-0.3%	-0.7%	-2.4%
Grand Total	0.5%	0.2%	0.2%	-0.2%	0.4%	-0.3%	0.4%	0.1%	0.3%	0.3%	-0.5%	-1.3%	0.0%

Assignment Validation

- Great fit!
 - Better than old model
 - Far exceeds TDOT standards
 - No ODME, only screenline factoring

VOLUME RANGE	RMSE	TDOT MAXIMUM
< 5,000	62.1%	100%
5,000 to 10,000	37.9%	45%
10,000 to 15,000	28.0%	35%
15,000 to 20,000	22.7%	30%
20,000 to 30,000	15.7%	27%
30,000 to 50,000	14.1%	25%
50,000 to 60,000	9.9%	20%
All	29.0%	45%

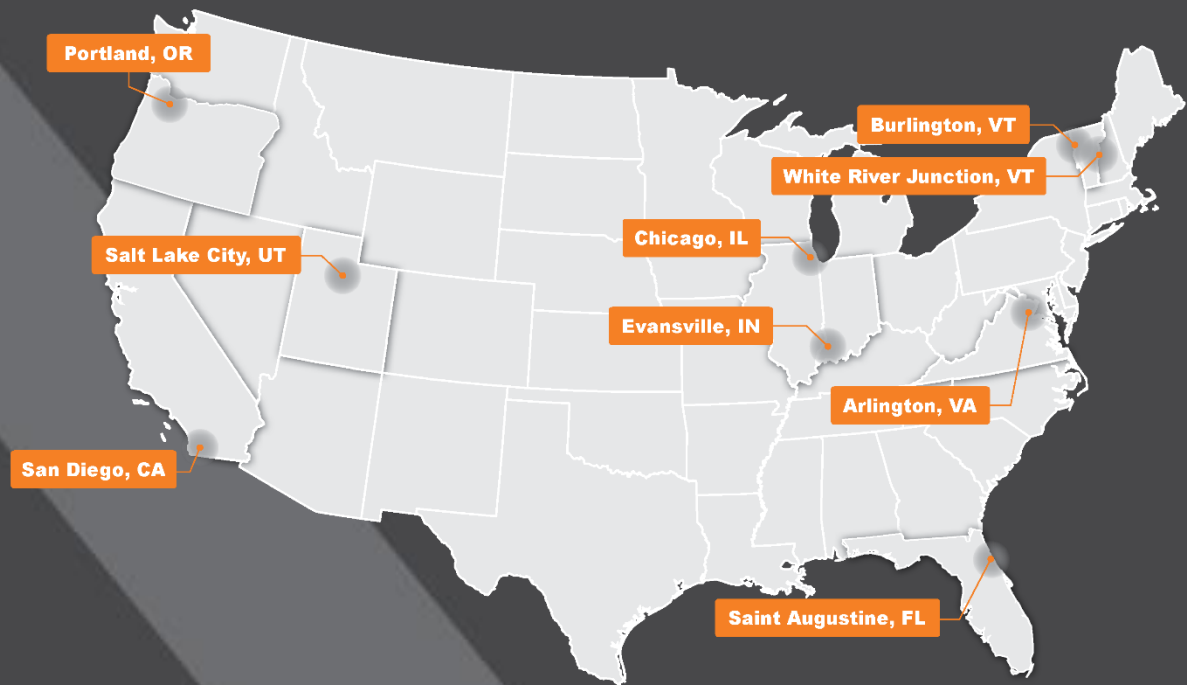




Final Thoughts

What's Next?

- Data Driven Forecasting
 - Pivoting, destination choice models with constants
 - Better accuracy, analog to STOPS
- Accelerating Pace of Change
 - Transformational changes
 - Big data may provide key in more frequent updates
- Evolving Data & Methods
 - New data sources entering the market
 - Data fusion: surveys & big data



Contact

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